

Table 2: Comparison Between LAN, MAN and WAN

Network	Coverage area	Bandwidth	Links	Cost	Specialities
LAN	Typically, 1km; over a building, an institution, etc	Low	Ethernet cable	Low	- Fully-private network - Shared media network - Can support 100% resource sharing
MAN	Typically, 100km; over a city, zonal district, etc	Medium	Coaxial cable, microwave link	Medium	- Zonal public network - Switched network
WAN	Typically, over 100km to 10,000km; over a country or province	Highest	Satellite links, telephonic links	Most expensive	- National public network - Switched network
Internet	Beyond 10,000km; over multiple countries, intercontinental (planets)	Highest	Logical connectivity using physical networks	Not so expensive	Logical connection across the globe

Message switching has a few limitations, as follows:

- When a message is long, junking of the link creates problems in communication.
- Due to limitation of buffer at intermediate nodes, a long message may result in either loss of data or buffer congestion.
- In a long message, if the link is junked, priority assigned transmission may not be possible.

These limitations are duly-solved with packet switching. In packet switching, messages or data are fragmented or divided into many parts, each called a packet. Each packet is enveloped with some overheads for packet numbering, error control, source address, destination address, start of frame, end of frame, priority assignment, synchronisation and so on, to make a frame. Packets/frames are made to traverse in the network from node to node, independently, to reach the destination from source via best possible/optimised shared link(s).

Packet switching is inherently fast switching. Due to packetisation (availability of packets that may move independently) and shared links between source and destination

(availability of parallel links), different packets may traverse in different parallel paths. This results in lower transmission time.

However, overhead bits of frames/packets require more transmission time over original data. Thus, tradeoffs exist between parallel transport with overhead bits. Fewer number of packets amounts to possibility of use of fewer number of more parallel links, causing slower transport. More number of packets means use of more parallel paths, resulting in faster transport. And more packets use more overhead bits, leading to more transmission time, and lesser fast transport.

How many packets should be used and what should be the optimal size of each packet to derive maximum benefit out of packetisation? No specific engineering theory exists to answer these questions. Packet sizes are standardised for different data networks by different professional bodies. Table 1 has details on the same.

Packet switching is used in all networks, as shown in Table 1. It is used to transport conventional data. Over time, there emerges a logical need to have a switching technique for all services, voice, video and data

as multimedia services. In packet switching, packets are variable in sizes and relatively long. Variable packet sizes require higher processing time at transmitter (for making of frames with formation of overheads) as well as receiver (for removing overheads, rearranging packets in order, as packets may go out of order for transport over different available parallel paths, error checking, etc) sides.

Voice and video are time-sensitive services. Accordingly, packet switching was modified as asynchronous transfer mode (ATM) cell switching to make it applicable to voice and video, besides data. In ATM cell switching, packet size is fixed at 53 bytes with 48 bytes of data and five bytes of overhead. Due to a fixed packet size, node processing delay is minimised.

Further, the short size of a cell in ATM gives the benefit of availing more parallel paths and faster transport. This makes ATM (often known as very fast packet switching) an appropriate switching technique for multimedia services.

Packet switching is implemented in two modes: virtual circuit services (VCS) for huge data and datagram service (DS) for low data.

In VCS, logical virtual connection is pre-established for data transport between source and destination. All data of the session between source and destination follows the virtual link. But unlike telephony, the pre-established link is sharable.

In DS, the packet moves independently over available links between source and destination.

The motivation behind data networking was to connect geographically-distributed computing resources (standalone computers, terminals, printers, scanners, etc) through communication links (that may be wires, cables, optical-fibre lines, wireless links, etc) with a view to meaningful exchange of information. There are two primary goals: resource sharing and load sharing, so that costly printers, software and computing